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1 # calculator.py
2
3 def add(a, b):
4     return a + b
5
6 def subtract(a, b):
7     return a - b
8
9 def multiply(a, b):
10    return a * b
11
12 def divide(a, b):
13    if b == 0:
14        return "Error! Division by zero is not
15        allowed."
16    return a / b
17
18 while True:
19    print("\n==== Calculator ====")
20    print("1. Addition")
21    print("2. Subtraction")
22    print("3. Multiplication")
23    print("4. Division")
24    print("5. Exit")
25
26    choice = input("Enter your choice (1-5): ")
27
28    if choice == "5":
29        print("Thank you for using the
30        calculator!")
31        break
32
33    if choice in ["1", "2", "3", "4"]:
34        try:
35            num1 = float(input("Enter first
36            number: "))
37            num2 = float(input("Enter second
38            number: "))
39
40            if choice == "1":
41                print("Result:", add(num1, num2))
42            elif choice == "2":
43                print("Result:", subtract(num1,
44                num2))
45            elif choice == "3":
46                print("Result:", multiply(num1,
47                num2))
48            elif choice == "4":
49                print("Result:", divide(num1,
50                num2))
51        except ValueError:
52            print("Invalid input! Please enter
53            numeric values.")
54    else:
55        print("Invalid choice! Please select
56        between 1 and 5.")
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==== Calculator ====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 1
Enter first number: 10
Enter second number: 20
Result: 30.0

==== Calculator ====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 2
Enter first number: 10
Enter second number: 5
Result: 5.0

==== Calculator ====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 3
Enter first number: 15
Enter second number: 4
Result: 60.0

==== Calculator ====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 4
Enter first number: 10
Enter second number:
Invalid input! Please enter numeric values.

==== Calculator ====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 4
Enter first number: 10
Enter second number: 5
Result: 2.0

==== Calculator ====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 5
Thank you for using the calculator!

=== Code Execution Successful ===
```